

Department Overview • Department Name: Department of Electrical Engineering • Faculty: Faculty of Information & Communication Technology (FICT) • Establishment: Founded in 2012. The graduate (MS) program was subsequently initiated in Spring 2014. • Accreditation: The undergraduate program is accredited by the Pakistan Engineering Council (PEC) under Level-II, indicating compliance with high-quality education standards. • Core Focus: Focuses heavily on Electric Power Systems Engineering and Telecommunication Engineering. The curriculum balances mathematical depth and engineering fundamentals with professional skills like personality grooming, public communication, entrepreneurship, and management. Vision and Mission Vision: To become a global leader in the field of Electrical Engineering having a pivotal role in the development and prosperity of society. Mission: To impart quality learning in the field of electrical engineering, promote excellence in research & innovation, and play a leading role in economic, social, and cultural development. Program Educational Objectives (PEOs) The department's educational goals are designed to produce professional engineers who will:

1. Demonstrate professional competence in electrical engineering through in-depth knowledge and skills, alongside contemporary interdisciplinary studies.
2. Engage in continual professional development through effective communication, teamwork, and lifelong learning.
3. Depict professional integrity, practice, and implement ethics and social responsibility to the global community.

Bachelor of Science (BS) Program

Admission Requirements • F.Sc (Pre-Engineering): Completed from any recognized board or equivalent with at least 60% marks. • Diploma of Associate Engineer (DAE): Completed in a relevant engineering field with at least 60% marks.

Degree Requirements • Total Credit Hours: 136 • Semester Credit Load: 15-18 credit hours per semester. • Minimum Academic Standing: CGPA $\geq$ 2.0 • Specialization Streams: Students can choose to specialize in either Telecommunication Engineering or Power Systems Engineering.

Curriculum Plan (First 4 Semesters Schema) Note: Credit hour listings follow the format (Theory + Lab).

1st Semester • HUM-163 | Functional English | Knowledge Area: Humanities | Credit Hours: 3+0 • MATHP-118 | Calculus and Analytical Geometry | Knowledge Area: Natural Sciences | Credit Hours: 3+0 • EE-112/L | Linear Circuit Analysis & Lab | Knowledge Area: Engineering Foundation | Credit Hours: 3+1 • CS-123 | Applications of ICT | Knowledge Area: Computer Sciences | Credit Hours: 2+1 • ENGG-217L | Engineering Drawing | Knowledge Area: Engineering Foundation | Credit Hours: 0+1 • PHY-205/L | Applied Physics & Lab | Knowledge Area: Natural Sciences | Credit Hours: 2+1 • MTH-XX | *Math Deficiency - I (if applicable)* | Knowledge Area: Natural Sciences | Credit Hours: 3+0 • Semester Total Credit Hours: 17

2nd Semester • MATHA-214 | Differential Equations | Knowledge Area: Natural Sciences | Credit Hours: 3+0 • XX | Natural Science Elective | Knowledge Area: Natural Sciences | Credit Hours: 3+0 • ELE-102L | Electrical Workshop Practice | Knowledge Area: Engineering Foundation | Credit Hours: 0+1 • HUM-XX | Pakistan Studies | Knowledge Area: Humanities | Credit Hours: 2+0 • CS-112/L | Computer Programming & Lab | Knowledge Area: Advanced Computer & IT |

Credit Hours: 3+1 • EE-208/L | Electronic Devices & Circuits & Lab | Knowledge Area: Engineering Foundation | Credit Hours: 3+1 • MTH-XX | *Math Deficiency II (if applicable)* | Knowledge Area: Natural Sciences | Credit Hours: 3+0 • Semester Total Credit Hours: 17

3rd Semester • MATHA-213 | Complex Variables and Transforms | Knowledge Area: Natural Sciences | Credit Hours: 3+0 • HUM-116 | Ideology and Constitution of Pakistan | Knowledge Area: Humanities | Credit Hours: 2+0 • EE-110/L | Digital Logic Design & Lab | Knowledge Area: Engineering Foundation | Credit Hours: 3+1 • CS214/L | Data Structures and Algorithm & Lab | Knowledge Area: Advanced Computer & IT | Credit Hours: 3+1 • ELE-201/L | Electrical Network Analysis & Lab | Knowledge Area: Engineering Foundation | Credit Hours: 3+1 • ENVIRON-103 | Occupational Health and Safety | Knowledge Area: Multidisciplinary Eng. | Credit Hours: 1+0 • Semester Total Credit Hours: 18

4th Semester • HUM-XX | Social Sciences Elective | Knowledge Area: Humanities | Credit Hours: 2+0 • MATHP-111 | Linear Algebra | Knowledge Area: Natural Sciences | Credit Hours: 3+0 • HUM-101 | Islamic Studies/Ethics | Knowledge Area: Humanities | Credit Hours: 2+0 • MATHA-217 | Probability Methods in Engineering | Knowledge Area: Engineering Foundation | Credit Hours: 3+0 • EE-261/L | Signals and Systems & Lab | Knowledge Area: Engineering Foundation | Credit Hours: 3+1 • EE-111 | Electromagnetic Field Theory | Knowledge Area: Engineering Foundation | Credit Hours: 3+0 • Semester Total Credit Hours: 17

Core Higher Semester Subjects (Semesters 5 to 8 Highlights) • 5th Semester: Electrical Machines (3+1), Microprocessors and Interfacing (3+1), Communication Systems (3+1), Civics and Community Engagement (2+0), MDEE Elective (3+0). Total: 17 Credits. • 6th Semester: Expository Writing (3+0), Linear Control Systems (3+1), Power Distribution and Utilization (3+1), Depth Elective (Core)-1 (3+1), Project Management (2+0). Total: 17 Credits. • 7th Semester: Depth Elective-II Core (3+1), Depth Elective-III (3+1), Flexible Elective I (Climate Change and Risk Management, 2+1), Depth Elective-IV (3+0), Arts & Humanities Elective (Muslim Heritage, 2+0), Final Year Design Project Part-I (0+2). Total: 18 Credits. • 8th Semester: Entrepreneurship (2+0), Depth Elective-V (3+0), Flexible Elective-II (3+0), Flexible Elective-III (3+0), Final Year Design Project Part-II (0+4). Total: 15 Credits.

Master of Science (MS) Program

Admission Requirements • Academic Background: 16 years of education or equivalent (such as a 4-year B.E or BS degree) in a relevant field from an HEC-recognized institution. • Minimum Grades: At least 60% marks under the annual system, or a minimum CGPA of 2.5 out of 4.0. • Testing: A valid GAT General score with at least 50% marks.

Degree Requirements • Total Credit Hours: 30 • Coursework: 24 credit hours of graduate-level courses (8 courses). • Research Thesis: 6 credit hours of research conducted under the supervision of PhD faculty members. • Minimum Passing Standing: CGPA $\geq$ 2.5 • Areas of Research: Specialized streams are available in Telecommunication Systems Engineering and Power Systems Engineering.

Future Program Plans The department is actively working toward launching a Ph.D. Engineering Program, which will broaden specialized research tracks across several fields: • Power Systems Engineering • Telecommunication Systems Engineering • Electronic Engineering • Control Systems Engineering • Computer Systems Engineering